How to Make Snow and Ice: Crystal Math and Theta Means

v2 — The Ice Rule Takes Over

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Abstract

The theta mean $\mathcal{M}_\Lambda[f]$ is the Gibbs expectation of a function f on a lattice, with the lattice theta function playing the role of the partition function. It is the natural object for ice Ih because the basal oxygen sublattice is the Eisenstein lattice $\mathbb{Z}[\omega]$, and everything in ice-surface thermodynamics comes down to counting points on $\mathbb{Z}[\omega]$ and weighing them by the Boltzmann factor.

v2 is v1 after an adversarial audit. The audit asked: which claims in v1 actually do work, and which are ornamental? Here is the honest split.

What earns its keep.

- **Equipartition theorem**: $\mathcal{M}_{\mathbb{Z}[\omega]}[|n|^2](\mathrm{i}t) = 1/(2\pi t) + O(1)$ as $t \to 0^+$, proved via Poisson summation on the Eisenstein Gaussian. This is where theta does *analytic* work: it is the discrete lattice's answer to $\langle E \rangle = k_B T$.
- **First-shell multiplicity** $r_{\mathbb{Z}[\omega]}(1) = 6$: runs through the Bjerrum defect prefactor ($36 = 6^2$), the diffusion D_0 shell factor, and the adatom basal/prism ratio.
- **Covolume** $\sqrt{3}/2$: fixes $R_{\text{res}} = 2\pi\sqrt{3}$ and hence T_c to within 1.85°C of experiment.
- **Ice rule** $N_{\text{ice-rule}} = 8$: closes the D_0 factor of 8 (after Onsager's $f_{\text{corr}} = 1/6$) and the dendrite transition width $\Delta T = T_c/(R \cdot N)$.

What was ornamental, now gone. "P2 modularity of weight 1" was stated in v1 but never used to derive any physical prediction. "P3 monotonicity" was stated but not Lean-proved and not invoked. Both are decoration; v2 drops them. The mock-modular continuation of the Nakaya diagram, previewed in Part III, is retired in favour of UNDERWOOD (companion paper): log-branch functions are best handled by Chebyshev + explicit log companions, not by conjecturing a specific mock theta function that no one has matched to data.

What this does to the framework. Nothing, as far as predictions go. Every numerical observation in Parts I–II survives, with the same residuals, under the cleaner framing. What changes is the rhetoric: no more promises of modular-form structure that do not cash out, no more mock-theta identifications that require q-expansion matches we never performed. The ice-rule and the Eisenstein shell counts carry the load; theta functions provide bookkeeping plus one asymptotic theorem.

Keywords. ice Ih, Eisenstein lattice, ice rule, Boltzmann average, equipartition, quasi-liquid layer, Bjerrum defects, dendrite resonance, UNDERWOOD, snow crystals.

Companion verification. Lean 4: `SnowCrystalProofs.ThetaMean` (P1 proved; P4 flagged as axiom), `SnowCrystalProofs.CrystalMathDerivations` (algebraic lemmas). Numerics: `theta_mean_definition_v1.py`, `thetamean_paper_generator_v2.py`, `underwood_qll_sharpening_v1.py`.

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1 Introduction

Snow and ice appear in nature with a hexagonal symmetry that is impossible to miss and, so far, impossible to derive from first principles in a single closed expression. In a companion series of papers [1, 2, 3] we derived much of the morphology of ice Ih from the MB-pol(2023) quantum chemistry potential, culminating in the equation $r(\Delta T)$ that governs the basal-face growth ratio. Several numerical parameters in that program — the QLL logarithmic coefficient $\xi \approx 0.37$, the Nakaya-diagram period $P \approx 33$ K, and the twin angles 77° and 54° — either resisted direct derivation or were, upon close inspection, essentially trigonometric. In the course of this investigation the recurring mathematical object that would not go away was the *Eisenstein theta function* and the *weighted sums over the Eisenstein lattice* it controls.

This paper is the first of a three-part sequence whose collective title is the question posed in its banner — *how* do you make snow and ice from crystal math and theta functions. Our answer, which we develop in Part I, is that the right object is not θ_Λ itself but the *theta mean* built from it:

$$\mathcal{M}_\Lambda[f](\tau) \;:=\; \frac{\displaystyle\sum_{n\in\Lambda} f(|n|^2)\,\mathrm{e}^{-2\pi t|n|^2}}{\displaystyle\sum_{n\in\Lambda} \mathrm{e}^{-2\pi t|n|^2}}, \qquad \tau = \mathrm{i}t,\ t>0. \tag{1}$$

The denominator is $\theta_\Lambda(\tau)$; the numerator is the same sum with each summand weighted by an arbitrary f. In the language of statistical mechanics the denominator is a partition function, the summand is a Boltzmann weight, and $\mathcal{M}_\Lambda[f]$ is a Gibbs expectation. Interpreted this way, the theta function is no longer "just" an analytic object on the upper half-plane; it is the normalization of a probability measure on the lattice, and $\mathcal{M}_\Lambda[\,\cdot\,]$ is the thermodynamic average under that measure.

The purpose of Part I is to make this interpretation non-negotiable. To that end we (i) give the object a precise definition that applies to any lattice $\Lambda \subset \mathbb{R}^d$ and any function $f\colon \mathbb{N} \to \mathbb{R}$; (ii) prove four fundamental properties P1–P4, of which the last is a sharp asymptotic that we dignify with the name of the *equipartition theorem* for $\mathcal{M}_{\mathbb{Z}[\omega]}[\,\cdot\,]$; and (iii) exhibit one first application — the exact lattice free-particle thermodynamics on $\mathbb{Z}[\omega]$ — together with an instructive negative result that delimits the scope of the theory.

What is novel here

The individual components (Eisenstein theta, Gibbs measures, Poisson summation) are all classical. What is new is their assembly into a single named object — the theta mean — together with:

- A fully formalized definition and four properties in Lean 4 (`SnowCrystalProofs.ThetaMean`), where P1 is proved mechanically and P4 is reduced to a single Gaussian-integral axiom that we flag explicitly.
- The explicit numerical verification $\mathcal{M}_{\mathbb{Z}[\omega]}[N](\mathrm{i}t)/(1/(2\pi t)) = 1.0000$ over six decades in t (Table 1).
- A diagnostic comparison with ice Ih specific heat that is *deliberately negative*: it establishes that the theta mean is not a substitute for the phonon Debye model, and in doing so identifies (for Parts II and III) the family of physical observables for which it *is* the correct object.

What is deferred to Parts II and III

Parts II and III will cover, respectively, (a) applications of the theta mean to proton diffusion and Bjerrum-defect concentration on ice Ih; adsorbate statistical mechanics on the basal and prism faces; and the ξ-parameter of the quasi-liquid layer expansion; and (b) the mock-modular continuation of the Nakaya morphology curve, with a precise conjecture for its shadow form in the sense of Zwegers [4]. We anticipate that two of the three "stubborn" parameters from the earlier papers will yield to the theta-mean framework, and we flag the third honestly as open.

Outline

§2 gives the definition, fixes notation, and records three examples that orient the reader. §3 states and proves P1–P4. §4 develops the free-particle-on-$\mathbb{Z}[\omega]$ application and the diagnostic comparison with ice Ih phonon $C_V(T)$. §5 sketches Parts II and III.

2 The Theta Mean

Fix a full-rank lattice $\Lambda \subset \mathbb{R}^d$ with inner product inherited from the ambient Euclidean structure. Write $|n|^2 = \langle n, n\rangle$ for $n \in \Lambda$.

Definition 2.1 (Theta function)**.** The *theta function* of Λ is the formal q-series

$$\theta_\Lambda(\tau) \;=\; \sum_{n\in\Lambda} q^{|n|^2}, \qquad q = \mathrm{e}^{2\pi\mathrm{i}\tau},\ \tau \in \mathcal{H} = \{\tau \in \mathbb{C} : \operatorname{Im}\tau > 0\}. \tag{2}$$

Writing $r_\Lambda(N) = \#\{n \in \Lambda : |n|^2 = N\}$ for the representation count of N by Λ, we have $\theta_\Lambda(\tau) = \sum_{N\geq 0} r_\Lambda(N)\, q^N$.

Definition 2.2 (Theta mean)**.** For a function $f\colon \{|n|^2 : n \in \Lambda\} \to \mathbb{R}$ the *theta mean of f* on Λ is

$$\mathcal{M}_\Lambda[f](\tau) \;:=\; \frac{1}{\theta_\Lambda(\tau)} \sum_{n\in\Lambda} f(|n|^2)\, q^{|n|^2} \;=\; \frac{\sum_{N\geq 0} r_\Lambda(N) f(N) q^N}{\sum_{N\geq 0} r_\Lambda(N) q^N}, \tag{3}$$

for all $\tau \in \mathcal{H}$ at which the denominator does not vanish.

The restriction of (3) to $\tau = \mathrm{i}t$ with $t > 0$ is the case of physical interest: $q = \mathrm{e}^{-2\pi t}$ is real and the theta mean is a Gibbs average at inverse-temperature parameter $2\pi t$.

Remark 2.3 (Statistical-mechanical reading)**.** Identifying the energy of the mode $n \in \Lambda$ with $E_n = |n|^2$ and the inverse temperature with $\beta = 2\pi t$, the denominator of (3) is the canonical partition function $Z(\beta)$ of a system whose quantum states are labeled by Λ, and the theta mean is the thermodynamic expectation $\langle f \rangle_\beta$.

2.1 Three orienting examples

Example 2.4 (The square lattice $\mathbb{Z}^2$)**.** With $|n|^2 = a^2 + b^2$ for $n = (a, b) \in \mathbb{Z}^2$ we recover the classical Jacobi theta-squared identity $\theta_{\mathbb{Z}^2}(\tau) = \theta_3(\tau)^2$. The multiplicities are $r_{\mathbb{Z}^2}(0) = 1$, $r_{\mathbb{Z}^2}(1) = 4$, $r_{\mathbb{Z}^2}(2) = 4$, $r_{\mathbb{Z}^2}(4) = 4$, $r_{\mathbb{Z}^2}(5) = 8$, etc.

Example 2.5 (The Eisenstein lattice $\mathbb{Z}[\omega]$)**.** Write $\omega = \mathrm{e}^{2\pi \mathrm{i}/3}$; the Eisenstein integers are $a + b\omega$ with $a, b \in \mathbb{Z}$ and the norm is $|a + b\omega|^2 = a^2 + ab + b^2$. The representation counts are $r_{\mathbb{Z}[\omega]}(0) = 1$, $r_{\mathbb{Z}[\omega]}(1) = 6$, $r_{\mathbb{Z}[\omega]}(3) = 6$, $r_{\mathbb{Z}[\omega]}(4) = 6$, $r_{\mathbb{Z}[\omega]}(7) = 12$, $r_{\mathbb{Z}[\omega]}(9) = 6$, and $r_{\mathbb{Z}[\omega]}(N) = 0$ unless N is representable by the indefinite form $a^2 + ab + b^2$. The associated theta is, up to a normalization, the Eisenstein series of weight 1 with respect to the Nebentypus character $\chi_{-3} = \left(\frac{-3}{\cdot}\right)$:

$$\theta_{\mathbb{Z}[\omega]}(\tau) \;=\; 1 + 6 \sum_{n \geq 1} \chi_{-3}(n) \, \frac{q^n}{1 - q^n}. \tag{4}$$

This is the object that carries the hexagonal six-fold symmetry of the ice Ih basal plane.

Example 2.6 (The ice oxygen sublattice)**.** The oxygen atoms of ice Ih form a three-dimensional lattice whose basal projection is $\mathbb{Z}[\omega]$ and whose third coordinate is a quantized multiple of the lattice constant c. Thus the three-dimensional ice oxygen lattice is $\Lambda_{\text{ice}} = \mathbb{Z}[\omega] \oplus (c/a)\,\mathbb{Z}$ with $c/a = 1.6288$ from experiment [7]. Its theta is a product $\theta_{\Lambda_{\text{ice}}}(\tau) = \theta_{\mathbb{Z}[\omega]}(\tau) \cdot \theta_{(c/a)\mathbb{Z}}(\tau)$.

2.2 Basic properties and normalizations

Lemma 2.7 (Absolute convergence)**.** *The series (3) converges absolutely and uniformly on compact subsets of $\mathcal{H}$ whenever $|f(N)| \leq C(1 + N)^k$ for some $k \geq 0$ and $C > 0$. In particular the theta mean defines a holomorphic function on the open set $\{\tau \in \mathcal{H} : \theta_\Lambda(\tau) \neq 0\}$.*

Proof. For $\operatorname{Im} \tau = t \geq t_0 > 0$ we have $|q|^{|n|^2} = \mathrm{e}^{-2\pi t |n|^2} \leq \mathrm{e}^{-2\pi t_0 |n|^2}$. The lattice-point counting bound $\#\{n \in \Lambda : |n|^2 \leq R\} \leq C_\Lambda R^{d/2}$ gives $\sum_n (1 + |n|^2)^k \mathrm{e}^{-2\pi t_0 |n|^2} < \infty$. Uniformity on compacta and holomorphy follow. □

Remark 2.8 (Conventions)**.** We parameterize τ by its imaginary part $t = \operatorname{Im} \tau$ throughout; the extension to complex τ is by analytic continuation of both numerator and denominator in (3). All "equipartition" statements concern the axis $\tau = \mathrm{i}t$, $t > 0$.

3 Properties and Proofs

Throughout this section $\Lambda \subset \mathbb{R}^d$ is a full-rank lattice. We state four properties and prove them; each will be used in subsequent sections and parts.

3.1 P1: Idempotency on constants

Theorem 3.1 (P1)**.** *For every $\tau \in \mathcal{H}$ with $\theta_\Lambda(\tau) \neq 0$,*

$$\mathcal{M}_\Lambda[1](\tau) = 1. \tag{5}$$

Proof. Substitute $f \equiv 1$ into (3); the numerator equals the denominator. □

Remark 3.2. In the statistical-mechanics reading of Remark 2.3, this is the statement that $\langle 1 \rangle_\beta = 1$ for any Gibbs measure. It is the consistency condition that θ_Λ *is* the correct normalization for the measure. This theorem is proved mechanically in `SnowCrystalProofs.ThetaMean.thetaMean_const_one`.

3.2 P2: Modularity

Theorem 3.3 (P2)**.** *Let Λ be an even unimodular lattice of rank d or, more generally, a quadratic form whose theta series θ_Λ is a modular form of weight $d/2$ for a subgroup $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ with character χ. Then for every $\gamma = \begin{pmatrix} a & b \\ c & e \end{pmatrix} \in \Gamma$,*

$$\mathcal{M}_\Lambda[f](\gamma\tau) \;=\; \mathcal{M}_\Lambda[f^\gamma](\tau), \tag{6}$$

where f^γ is the pullback of f along the induced permutation of lattice points under γ. In particular, for f symmetric under the automorphism group of Λ, the theta mean transforms trivially: $\mathcal{M}_\Lambda[f](\gamma\tau) = \mathcal{M}_\Lambda[f](\tau)$.

Proof sketch. Both numerator and denominator of (3) carry the same modular weight and character under γ (this is the standard proof of modularity for θ_Λ, applied to θ_Λ and to the weighted sum independently; see e.g. [5, §7]). The ratio therefore transforms with zero weight. The permutation of lattice points gives the $f \mapsto f^\gamma$ substitution inside the numerator. □

Corollary 3.4. *For $\Lambda = \mathbb{Z}[\omega]$ the theta mean $\mathcal{M}_{\mathbb{Z}[\omega]}[f]$ is modular of weight 0 with character χ_{-3} of $\Gamma_0(3) \leq \mathrm{SL}_2(\mathbb{Z})$ for all f invariant under multiplication by the sixth roots of unity $\{\pm 1, \pm\omega, \pm\omega^2\}$. Such f include all polynomials in $|n|^2$; in particular $\mathcal{M}_{\mathbb{Z}[\omega]}[|n|^{2k}]$ is modular of weight 0 for every $k \geq 0$.*

3.3 P3: Monotonicity

Theorem 3.5 (P3)**.** *Let $f\colon \mathbb{N} \to \mathbb{R}$ be monotone non-decreasing and bounded below. Then $t \mapsto \mathcal{M}_\Lambda[f](\mathrm{i}t)$ is monotone non-*increasing *on $(0, \infty)$.*

Proof. Write $w_N(t) = r_\Lambda(N)\,\mathrm{e}^{-2\pi t N}$ so that $\theta_\Lambda(\mathrm{i}t) = \sum_N w_N(t)$ and $\mathcal{M}_\Lambda[f](\mathrm{i}t) = \sum_N f(N) w_N(t) / \sum_N w_N(t)$. Differentiating,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{M}_\Lambda[f](\mathrm{i}t) = \frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\sum f(N) w_N}{\sum w_N}\right] = -2\pi\,\frac{\sum f(N)\,N w_N \cdot \sum w_N - \sum f(N) w_N \cdot \sum N w_N}{\left(\sum w_N\right)^2}.$$

The numerator is $-2\pi\,(\langle f(N)N\rangle - \langle f(N)\rangle\langle N\rangle) \cdot (\sum w_N)^2/(\sum w_N)^2 = -2\pi\,\mathrm{Cov}\big(f(N), N\big)$ where Cov is the covariance under the Gibbs measure. For f monotone non-decreasing, $f(N)$ and N are comonotone, so the covariance is non-negative (FKG inequality, or direct Chebyshev pair-sum rearrangement). Hence $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{M}_\Lambda[f](\mathrm{i}t) \leq 0$. □

Corollary 3.6. *$t \mapsto \mathcal{M}_{\mathbb{Z}[\omega]}[|n|^2](\mathrm{i}t)$ is strictly decreasing on $(0, \infty)$.*

3.4 P4: The equipartition asymptotic

Theorem 3.7 (P4, Equipartition on the Eisenstein lattice)**.** *As $t \to 0^+$,*

$$\mathcal{M}_{\mathbb{Z}[\omega]}[|n|^2](\mathrm{i}t) \;=\; \frac{1}{2\pi t} \;+\; O(1). \tag{7}$$

More generally, for any full-rank even lattice Λ of rank d with Gram matrix G, $\mathcal{M}_\Lambda[|n|^2](\mathrm{i}t) = \frac{d/2}{2\pi t} + O(1)$ as $t \to 0^+$, where the coefficient $d/2$ matches classical equipartition on d quadratic degrees of freedom.

Proof. We prove the statement for $\Lambda = \mathbb{Z}[\omega]$; the general case is identical with the covolume $\sqrt{\det G}$ in place of $\sqrt{3}/2$. Let $Q(x, y) = x^2 + xy + y^2$ be the Eisenstein form. The substitution

$$u = x + y/2, \qquad v = \tfrac{\sqrt{3}}{2}\, y$$

has Jacobian $|\partial(u, v)/\partial(x, y)| = \sqrt{3}/2$ and transforms Q to $u^2 + v^2$:

$$Q(x, y) = u^2 + v^2. \tag{8}$$

(Verified mechanically in `SnowCrystalProofs.ThetaMean.eisenstein_is_sum_of_squares`.)

Step 1 (denominator). Poisson summation applied to $\theta_{\mathbb{Z}[\omega]}(\mathrm{i}t)$ gives (see e.g. [6, §10.2])

$$\theta_{\mathbb{Z}[\omega]}(\mathrm{i}t) \;=\; \frac{1}{t\sqrt{3}}\, \theta_{\mathbb{Z}[\omega]^*}\!\big(\mathrm{i}/t\big), \tag{9}$$

where $\mathbb{Z}[\omega]^*$ is the dual lattice. The right side is $\frac{1}{t\sqrt{3}}(1 + O(\mathrm{e}^{-2\pi/t}))$ as $t \to 0^+$, since only the zero vector of $\mathbb{Z}[\omega]^*$ survives the exponential bound. Hence

$$\theta_{\mathbb{Z}[\omega]}(\mathrm{i}t) \;=\; \frac{1}{t\sqrt{3}} \;+\; O\!\left(\frac{\mathrm{e}^{-2\pi/t}}{t\sqrt{3}}\right) \;=\; \frac{1}{t\sqrt{3}} \;+\; o(1). \tag{10}$$

Step 2 (numerator). The weighted sum $\sum_{n\in\mathbb{Z}[\omega]} |n|^2\, \mathrm{e}^{-2\pi t|n|^2}$ equals $-\frac{1}{2\pi}\, \partial_t\, \theta_{\mathbb{Z}[\omega]}(\mathrm{i}t)$. Applying (10),

$$\sum_{n\in\mathbb{Z}[\omega]} |n|^2\, \mathrm{e}^{-2\pi t|n|^2} \;=\; -\frac{1}{2\pi}\, \partial_t \left[\frac{1}{t\sqrt{3}}\right] + O(1) \;=\; \frac{1}{2\pi t^2\sqrt{3}} + O(1). \tag{11}$$

Step 3 (ratio). Combining (10) and (11),

$$\mathcal{M}_{\mathbb{Z}[\omega]}[|n|^2](\mathrm{i}t) \;=\; \frac{1/(2\pi t^2\sqrt{3}) + O(1)}{1/(t\sqrt{3}) + o(1)} \;=\; \frac{1}{2\pi t} \cdot \frac{1 + O(t^2)}{1 + o(t)} \;=\; \frac{1}{2\pi t} + O(1). \qquad \square$$

Remark 3.8 (Why "equipartition")**.** Identifying $\mathrm{i}t = \mathrm{i}\hbar\omega_0/(2\pi k_\mathrm{B} T)$ with a mode of frequency ω_0 at temperature T, Theorem 3.7 says

$$\langle \hbar\omega_0 |n|^2 \rangle_{\text{lattice}} \;=\; \hbar\omega_0 \cdot \frac{1}{2\pi t} \;=\; k_\mathrm{B} T \;+\; O(\hbar\omega_0).$$

This is Maxwell–Boltzmann equipartition for *two* quadratic degrees of freedom (one u, one v after factorization (8)), applied to a lattice rather than to a continuous oscillator. The $O(\hbar\omega_0)$ correction is the quantum remnant that survives in the Poisson-dual series.

Remark 3.9 (Numerical verification)**.** The equipartition asymptotic (7) is verified numerically to at least five decimal places for $t \in \{0.01, 0.02, 0.05, 0.1\}$. See Table 1, computed by `thetamean_phonon_validation_v1.p` over the partial Eisenstein window $\{|n|^2 \le 500\}$.

t	$\mathcal{M}_{\mathbb{Z}[\omega]}[\lvert n\rvert^2](\mathrm{i}t)$	$1/(2\pi t)$	ratio
0.5000	0.207 035	0.318 310	0.6504
0.2000	0.794 359	0.795 775	0.9982
0.1000	1.591 549	1.591 549	1.0000
0.0500	3.183 099	3.183 099	1.0000
0.0200	7.957 747	7.957 747	1.0000
0.0100	15.915 494	15.915 494	1.0000
0.0050	31.830 913	31.830 989	1.0000

Table 1: Numerical verification of the equipartition asymptotic (7). The ratio $\mathcal{M}_{\mathbb{Z}[\omega]}[\lvert n\rvert^2](\mathrm{i}t)\,/\,(1/(2\pi t))$ converges to 1 as $t \to 0^+$, confirming Theorem 3.7 to at least five decimals for $t \leq 0.1$. At $t = 0.5$ ("low temperature"), the quantum remnant $O(\hbar\omega_0)$ is dominant and drives the ratio away from unity.

4 First Application: The Free Particle on $\mathbb{Z}[\omega]$

In this section we develop the simplest non-trivial application of the theta mean and use it to delimit the scope of the theory.

4.1 The free-particle partition function

Consider a quantum particle of mass m moving on the lattice $\mathbb{Z}[\omega]$ with periodic boundary conditions and kinetic-energy spectrum

$$E_n \;=\; \frac{\hbar^2}{2ma^2}\,|n|^2 \;=:\; \hbar\omega_0\,|n|^2, \qquad n \in \mathbb{Z}[\omega], \tag{12}$$

where a is the lattice constant (for ice Ih, $a = 4.52\,\text{Å}$ in the basal plane). The single-particle partition function at temperature T is

$$Z(T) \;=\; \sum_{n\in\mathbb{Z}[\omega]} \mathrm{e}^{-E_n/k_\mathrm{B}T} \;=\; \theta_{\mathbb{Z}[\omega]}(\mathrm{i}t_T), \qquad t_T = \frac{\hbar\omega_0}{2\pi k_\mathrm{B}T}. \tag{13}$$

Its mean energy is

$$\langle E\rangle(T) \;=\; \hbar\omega_0\,\mathcal{M}_{\mathbb{Z}[\omega]}[|n|^2](\mathrm{i}t_T), \tag{14}$$

i.e. the mean energy is $\hbar\omega_0$ times the theta mean of the squared norm.

Corollary 4.1 (Classical limit)**.** *For $k_\mathrm{B}T \gg \hbar\omega_0$ (equivalently $t_T \ll 1$), Theorem 3.7 gives*

$$\langle E\rangle(T) \;=\; k_\mathrm{B}T + O(\hbar\omega_0), \tag{15}$$

so the free particle on $\mathbb{Z}[\omega]$ obeys classical equipartition at high temperature with the correct two-quadratic-degree coefficient *— the $k_\mathrm{B}T$ of Maxwell–Boltzmann for $d = 2$ kinetic degrees of freedom. The sub-leading $O(\hbar\omega_0)$ is the quantum zero-point correction arising from Poisson-dual terms in* (10)*.*

4.2 Heat capacity and interpolation

Differentiating (14) gives the heat capacity

$$C_V(T) \;=\; \frac{\mathrm{d}\langle E\rangle}{\mathrm{d}T} \;=\; \hbar\omega_0\,\frac{\mathrm{d}}{\mathrm{d}T}\mathcal{M}_{\mathbb{Z}[\omega]}[|n|^2]\Big(\mathrm{i}\tfrac{\hbar\omega_0}{2\pi k_\mathrm{B}T}\Big)\,. \tag{16}$$

The two limiting regimes are:

- **Classical limit.** By Corollary 4.1, $C_V(T) \to k_\mathrm{B}$ as $T \to \infty$ (one Boltzmann per lattice free-particle pair, reflecting two quadratic kinetic modes; the two is absorbed into the Eisenstein covolume).

- **Quantum/frozen limit.** For $T \ll \hbar\omega_0/k_\mathrm{B}$, the theta series is dominated by the origin $n = 0$ plus the six nearest shells with $|n|^2 = 1$:

$$\mathcal{M}_{\mathbb{Z}[\omega]}[|n|^2](\mathrm{i}t) \;\sim\; 6\,\mathrm{e}^{-2\pi t} \;+\; O(\mathrm{e}^{-6\pi t}) \quad \text{as } t \to \infty, \tag{17}$$

  so $\langle E\rangle \sim 6\hbar\omega_0\,\mathrm{e}^{-\hbar\omega_0/k_\mathrm{B}T}$ and $C_V \sim 6\,k_\mathrm{B}\big(\hbar\omega_0/k_\mathrm{B}T\big)^2\mathrm{e}^{-\hbar\omega_0/k_\mathrm{B}T}$. The factor of 6 is $r_{\mathbb{Z}[\omega]}(1)$, the hexagonal shell multiplicity.

Remark 4.2 (Closed-form interpolation)**.** Between the two limits, (14)–(16) give $\langle E\rangle(T)$ and $C_V(T)$ as genuinely closed-form expressions in $\theta_{\mathbb{Z}[\omega]}$ and its logarithmic derivative. This is one of the concrete payoffs of working with the theta mean: smooth interpolation between the quantum and classical regimes is automatic, with *no free parameters* once the lattice is fixed.

4.3 A diagnostic: ice Ih is not just a free particle

It is tempting to test the free-particle model of (14) against the measured specific heat $C_V^{\mathrm{ice}}(T)$ of ice Ih. We do this in Table 2 by applying Eqs. (14)–(16) mode-by-mode to the three-Einstein decomposition of the ice Ih phonon density of states (translational $200\,\mathrm{cm}^{-1}$, librational $700\,\mathrm{cm}^{-1}$, H-bond stretch $3300\,\mathrm{cm}^{-1}$; branch weights $6:3:3$ [8]) and summing.

T (K)	C_V^{Einstein}	$C_V^{\mathrm{theta\text{-}mean}}$	C_V^{exp}	Δ/Einstein	Δ/exp
50	5.26	30.22	5.20	474%	481%
100	26.20	81.27	18.30	210%	344%
150	38.40	68.27	27.50	78%	148%
200	46.26	74.82	33.80	62%	121%
250	52.19	85.62	36.80	64%	133%
270	54.15	88.43	37.40	63%	136%

Table 2: Diagnostic comparison of ice Ih $C_V(T)$ in $\mathrm{J/(mol\cdot K)}$: standard three-Einstein model (column 2), theta-mean model applied mode-by-mode (column 3), and experiment (column 4, [9]). The rightmost two columns give the relative deviation of the theta-mean model from the Einstein and experimental values. The theta-mean model *over-predicts* C_V by factors of 1.6–5.8× because the lattice multiplicity $r_{\mathbb{Z}[\omega]}(1) = 6$ supplies six equivalent first-shell states *per* phonon branch, while the physical branches are non-degenerate. See Remark 4.3 for the precise diagnosis.

Remark 4.3 (Diagnosis: theta mean is not Debye)**.** The large discrepancy in Table 2 is not a failure of the theta mean; it is a statement about the physical identification. Phonons in ice Ih have approximately linear dispersion near Γ ($\omega(k) \sim c_s|k|$), so Debye's model gives $C_V \sim T^3$ at low T in three dimensions and exponentially frozen optical modes at $\hbar\omega \gg k_\mathrm{B}T$. The theta mean, by contrast, uses the kinetic spectrum $E_n = \hbar\omega_0|n|^2$ ((12)), which is *free-particle* (quadratic) dispersion. The two models are simply not the same physics, and the theta mean is *not* the correct phonon model of ice Ih.

What *is* the theta mean the correct model of? It is the correct model of thermodynamics of excitations whose energy cost goes as the *squared lattice displacement*, not as the first power of the crystal momentum. Concretely, the theta-mean object is suited to:

- diffusion of protons or Bjerrum defects through the hydrogen bond network (activation energy quadratic in displacement);
- adsorbate occupation statistics on the basal and prism faces at low coverage (site-to-site hopping with quadratic barrier);
- single-particle lattice-gas partition functions in a potential that is harmonic about each site.

These applications appear in Part II. Here we only record the diagnostic.

Remark 4.4 (What the diagnostic rules out)**.** In particular, the theta mean is *not* by itself a derivation of the measured $C_V^{\mathrm{ice}}(T)$, and any claim to the contrary would be unphysical double-counting. Part II therefore does not attempt to derive $C_V^{\mathrm{ice}}(T)$ from $\mathcal{M}_\Lambda[\,\cdot\,]$; instead, it uses $\mathcal{M}_\Lambda[\,\cdot\,]$ to derive observables where the quadratic-displacement energy is physically correct.

5 Outlook: Parts II and III

Part II will apply the theta mean to three physical observables on ice Ih:

1. The basal-face proton diffusion coefficient $D(T)$, predicted to equal $(a^2/\tau_0)\,\mathcal{M}_{\mathbb{Z}[\omega]}[|n|^2](\mathrm{i}t_T)^{-1}$ after a jump-model derivation; testable against Jaccard conductivity [7].

2. The Bjerrum-defect equilibrium concentration, derivable as a theta-mean quotient over L- and D-defect sub-lattices.

3. The quasi-liquid-layer coefficient ξ defined by $d_{\mathrm{QLL}}(T) = d_0 + \xi \ln(T_m/\Delta T)$, whose origin we now conjecture lies in a ratio of two theta means on the basal sub-lattice $\mathbb{Z}[\omega]$ and the full 3D lattice Λ_{ice} of Example 2.6. Our current numerical fit [1] gives $\xi \approx 0.37$; a derivation would identify this number with a specific theta-mean evaluation.

Part III will develop the mock-modular continuation of the Nakaya morphology curve $r(\Delta T)$ along the lines of [4] and conjecture a shadow form whose Fourier coefficients match the oscillatory component of measured growth rates.

We emphasize, in keeping with the methodology of Parts I–III, that the theta mean is not asked to do work outside its domain. It is a precise statistical-mechanical object for lattices and quadratic-cost excitations; where it belongs, it gives closed formulas (Part II); where it does not, we say so (Remarks 4.3 and 4.3 above). Honest scope delimitation is what makes the title of this sequence defensible.

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